Enhancing LLM Translation Performance for *Spanish-Valencian* through Supervised Fine-tuning and Reinforcement Learning

Towards better fluency, naturalness and Low-Resource Dialect Adaptation

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84.68	62.16	20.63	0.544	41.0%	0.936
BEST CHRF ↑	BEST BLEU ↑	BEST TER ↓	BEST BLEURT ↑	MAX VA DIALECTAL ↑	BEST COMET ↑
+15.7 vs baseline	+22.9 vs baseline	−19.7 vs baseline	+0.287 vs baseline	+37.8 pp vs baseline	+0.03 vs baseline

1 Motivation & Research Questions

We explore three strategies to improve translation on `google/translategemma-4b-it`, a 4-billion-parameter translation-specialized LLM. The following approaches are adapted:

- **SFT**
- **SFT + GRPO v1**
- **SFT + GRPO v2**

Valencian, a Western Catalan dialect, has unique morphological and lexical traits that standard multilingual models often confuse or conflate with Eastern Catalan.

⚠ **Key Challenge:** TranslateGemma uses `target_lang_code="ca"` for both Catalan and Valencian, so it lacks dialect awareness.

1. *Can SFT on a small corpus significantly improve translation quality over a zero-shot multilingual baseline?*
2. *Does GRPO reinforcement learning with translation-quality reward signals further improve over SFT alone?*
3. *Can SFT and RL steer a model trained on standard Catalan towards producing Valencian-specific morphological and lexical forms?*
4. *Can naturalness of machine-generated translations be improved by incorporating human translation likeness signals into the GRPO reward function? Or would a reward based on automatic metrics be more effective?*

2 Training Pipeline

PIPELINE A

translategemma-4b-it baseline → SFT 50k pairs → GRPO v1 clf (HT vs. MT) · 5k

PIPELINE B ★ BEST

translategemma-4b-it baseline → SFT 50k pairs → GRPO v2 composite reward · 10k

3 HT/MT Classifier (part of the GRPOv1 reward)

We fine-tune `roberta-base-ca` (PlanTL-GOB-ES) as a binary classifier to distinguish **human translations (HT)** from **machine translations (MT)**.

TRAINING CORPORA

Corpus	Domain
TildeMODEL	General
DOGC	Official/Legal
Europarl	Parliamentary

MT SYSTEMS USED

System	Share
Helsinki opus-mt-es-ca	50%
NLLB-200-600M	50%

Train: 108k | Val: 12k

74.7	74.5
Accuracy	F1
74.7	75.2
Recall	Precision

Output: **P(HT|text) → r_t**. Test set is in Valencian (not standard Catalan), so some accuracy degradation is expected. Published as `guerreropaula/ht_mt_classifier_best`.

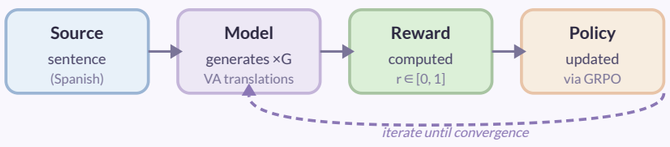
4 SFT

Base model	google/translategemma-4b-it
Trainable params	32.8M / 4.33B (0.76%)
SFT dataset	gplsi/amic_parallel — 50,000 pairs
Target modules	All attention and MLP layers
Quantization	4-bit NF4 + LoRA r=16 α=32

SFT Data: `gplsi/amic_parallel` — ES–VA domain-specific news from AMIC (Associació de Mitjans d'Informació i Comunicació).

5 GRPO

Model	SFT-Trained Model (TranslateGemma-4B-IT)



GRPO Data: gplsi/amic_parallel on Spanish source only. Valencian references are used to compute the reward signal.

Reward functions

GRPO v1 — chrF + Classifier

$$r = (1 - \alpha)\cdot r_c + \alpha\cdot r_t$$

- **r_c** = chrF / 100 (fidelity to the reference translation)
- **r_t** = P(HT|text) (naturalness)

Samples	5k
Steps	100 steps · 4 generations
Warm-up	Linear, 0.3 after step 50
Learning rate	5e-6
β	0.04

Best checkpoint: **step 80**

α increases **linearly** from 0 to **0.3** after a **50-step warm-up**, gradually introducing the naturalness signal.

GRPO v2 — Composite Reward ★ BEST

$$r = 0.5\cdot chrF + 0.3\cdot COMET + 0.2\cdot TTR - copy$$

- **chrF** = character-level fidelity to reference
- **COMET** = semantic quality estimate
- **TTR** = type-token ratio (measures lexical diversity)
- **copy** = −1 if source sentence is copied as output

Samples	10k
Steps	200 steps · 4 generations
β (KL penalty)	0.04
ε (PPO clip)	0.2
Learning rate	5e-6

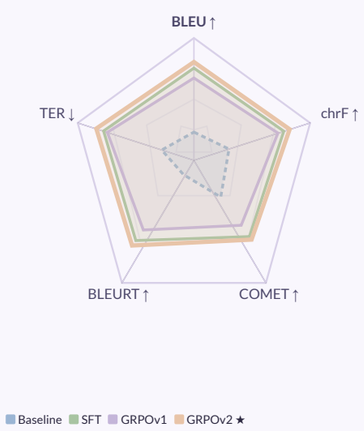
Best checkpoint: **step 100** (reward 0.831)

5 Main Results

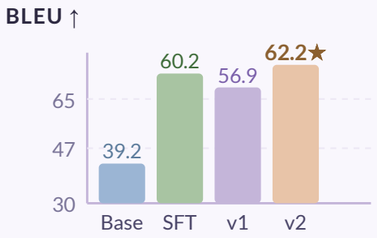
MODEL	CHRF ↑	BLEU ↑	TER ↓	BLEURT ↑	COMET ↑
Baseline	69.02	39.22	40.30	0.258	0.906
SFT	83.16	60.16	22.80	0.524	0.934
GRPO v1 (clf)	81.65	56.94	23.96	0.481	0.926
GRPOv2 ★	84.68	62.16	20.63	0.544	0.936

★ GRPOv2 best on all primary metrics. Test: `gplsi/ES-VA_translation_test` (1k)

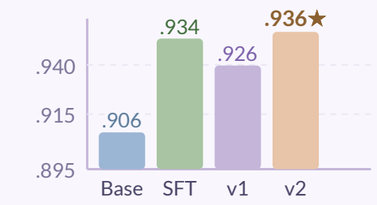
MULTIDIMENSIONAL EVALUATION



BLEU ↑



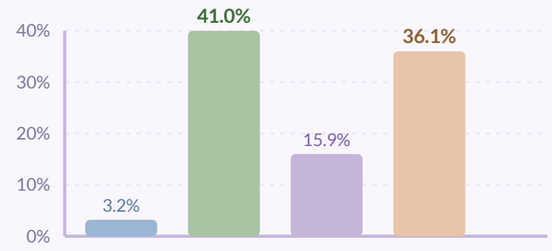
COMET ↑



6 Dialectal Valencian Analysis

We measure how often each model produces Valencian-specific morpho-lexical forms vs. Eastern Catalan equivalents, using a **30-feature vocabulary list**.

DIALECTAL VA SCORE (% VALENCIAN FORMS)



Representative features:

CA FORM	VA FORM	BASE	SFT	GRPOV2	GRPOV1
petit	***xicotet***	0%	100%	100%	0%
veure	***vore***	0%	83%	17%	0%
constitueixen	***constituïxen***	0%	0%	0%	0%
avui	***hui***	0%	71%	86%	0%
seva	***seua***	0%	100%	100%	45%

GRPO v1's HT/MT classifier trained on standard Catalan may penalize VA forms.

7 Qualitative Example

Source (ES): *"Su participación en este concurso le dio la oportunidad de entrar al mundo artístico."*

BASELINE

La seva participació en aquest concurs li va donar l'oportunitat d'entrar al món artístic.

chrF **76.6** BLEU **45.8** TER **21.4** BLEURT **0.465** COMET **0.953** TTR **1.0**

SFT

La seua participació en aquest concurs li va donar l'oportunitat d'ingressar al món artístic.

chrF **92.1** BLEU **80.0** TER **7.1** BLEURT **0.796** COMET **0.969** TTR **1.0**

GRPO V1

La seva participació en aquest concurs li va donar l'oportunitat d'ingressar al món artístic.

chrF **87.9** BLEU **68.7** TER **14.3** BLEURT **0.742** COMET **0.962** TTR **1.0**

GRPO V2 ★

La seua participació en aquest concurs li va donar l'oportunitat d'ingressar al món artístic.

chrF **92.1** BLEU **80.0** TER **7.1** BLEURT **0.796** COMET **0.969** TTR **1.0**

Valencian *seua* (green) vs Catalan *seva* (red)

8 Key Findings

1. **SFT achieves the largest improvement:** +14.1 chrF, +21.0 BLEU, −17.5 TER over zero-shot. Dialectal VA score also rises from 3.2% to 41.0%.
2. **GRPO v2 improves over SFT on every automatic metric**, while GRPO v1 actually regresses on all of them.
3. Despite its complexity, the **composite reward function outperforms the v1 chrF+classifier** on all automatic metrics.
4. **The classifier is the bottleneck for dialect.** Trained on standard Catalan, the HT/MT classifier may penalize Valencian morphology, which limits GRPO v1's performance.
5. **The GRPO v2 model adapts more fluent and natural structures**, as stated by BLEURT (+0.287 vs. baseline) and COMET (+0.03 vs. baseline).

9 Future Work

CLASSIFIER — Train an HT/MT classifier exclusively on Valencian HT data to prevent penalizing VA forms. 1

MORE SFT DATA — Augment SFT training samples beyond 50k since it was the most effective method for itself. 2

HUMAN EVAL — Native Valencian speaker (or ideally ES-VA translators) evaluation would provide important insights beyond automatic metrics. 3

REFINE REWARD FUNCTION — Desing a better aligned reward function and classifier tailored for naturalness. 4

10 Limitations

Limited coverage: Experiments are limited to a single language pair and domain (ES-VA news). Broader generalization across domains and language pairs remains untested.

Naturalness metrics: Automatic naturalness signals capture only lexical diversity. Writing style is broader and difficult to measure automatically.

Platform constraints: Running experiments on Google Colab limits GPU availability and training time.

Data scarcity: Limited ES-VA parallel data. Data augmentation could be considered, but Valencian references should consist of professional translations.

HUGGINGFACE MODELS

SFT
- `translategemma4b-sft-es-va2`

GRPO V1
- `80translategemma4b-grpo-es-va`

GRPO V2
- `translategemma4b-grpo2-es-va ⋆BEST`

HT/MT CLASSIFIER
- `ht_mt_classifier_best`

REFERENCES

Huiyuan Lai, Esther Ploeger, Rik Van Noord, and Antonio Toral. 2025. *Multi-perspective Alignment for Increasing Naturalness in Neural Machine Translation.* ACL 2025 (Long Papers), pp. 28071–28084.

Mao Zheng, Zheng Li, Yang Du, Bingxin Qu, and Mingyang Song. 2025. *Shy-hunyuan-MT at WMT25 General Machine Translation Shared Task.* 10th Conference on Machine Translation, pp. 607–613.

DATA & RESOURCES

gplsi/amic_parallel · gplsi/ES-VA_translation_test · Softcatalà parallel catalan corpus

GITHUB REPOSITORY